Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2015 – Paper 2 – Solutions**

June 2015 – Question 1 – Solution

(a) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be any two propositions.

(i) State the inverse and the contrapositive of the statement $\boldsymbol{p} \rightarrow \boldsymbol{q}$. [2]

Inverse : $\sim p \rightarrow \sim q$

Contrapositive : $\sim q \rightarrow \sim p$

(ii) Copy and complete the table below to show the truth table for

$\boldsymbol{p} \rightarrow \boldsymbol{q}$ and $\sim\boldsymbol{q} \rightarrow \sim\boldsymbol{p}$. [4]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{p}$	$\sim\boldsymbol{q}$	$\boldsymbol{p} \rightarrow \boldsymbol{q}$	$\sim\boldsymbol{q} \rightarrow \sim\boldsymbol{p}$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

(iii) Hence, state whether the compound statements $\boldsymbol{p} \rightarrow \boldsymbol{q}$ and $\sim\boldsymbol{q} \rightarrow \sim\boldsymbol{p}$ are logically equivalent. **Justify your response**. [2]

The compound statements $\boldsymbol{p} \rightarrow \boldsymbol{q}$ and $\sim\boldsymbol{q} \rightarrow \sim\boldsymbol{p}$ are logically equivalent since their truth values are the same as seen from Columns 5 and 6 in the table in (a)(ii).

(b) The polynomial $f(x) = x^3 + px^2 - x + q$ has a factor $(x - 5)$ and a remainder of 24 when divided by $(x - 1)$.

(i) Find the values of p and q. [4]

$f(x) = x^3 + px^2 - x + q$

Since $(x - 5)$ is a factor of $f(x)$, then by the Factor Theorem, $f(5) = 0$.

$f(5) = (5)^3 + p(5)^2 - (5) + q$

$0 = 125 + 25p - 5 + q$

$0 = 120 + 25p + q$

$25p + q = -120$ → Equation 1

Since $f(x)$ has a remainder of 24 when divided by $(x - 1)$, then by the Remainder Theorem, $f(1) = 24$.

$f(1) = (1)^3 + p(1)^2 - (1) + q$

$24 = 1 + p - 1 + q$

$q = 24 - p$ → Equation 2

Substituting Equation 2 into Equation 1 gives,

$25p + (24 - p) = -120$

$24p = -120 - 24$

$24p = -144$

$p = \frac{-144}{24}$

$p = -6$

Substituting $p = -6$ into Equation 2 gives,

$$q = 24 - (-6)$$

$$= 24 + 6$$

$$= 30$$

$\therefore$ The values of p and q are $p = -6$ and $q = 30$.

(ii) Hence, factorize $f(x) = x^3 + px^2 - x + q$ completely. [5]

$f(x) = x^3 + px^2 - x + q$

Since $p = -6$ and $q = 30$, then $f(x) = x^3 - 6x^2 - x + 30$.

We are given that $(x - 5)$ is a factor of $f(x)$.

By long division,

$$\begin{array}{r|l}
 & x^2 - x - 6 \\
\hline
x - 5 & x^3 - 6x^2 - x + 30 \\
 & x^3 - 5x^2 \\
\cline{2-2}
 & -x^2 - x \\
 & -x^2 + 5x \\
\cline{2-2}
 & -6x + 30 \\
 & -6x + 30 \\
\cline{2-2}
 & 0 \\
\cline{2-2}
\end{array}$$

So, we have,

$$f(x) = (x - 5)(x^2 - x - 6)$$

$$= (x - 5)(x - 3)(x + 2)$$

(c) Given that $S(n) = 5 + 5^2 + 5^3 + 5^4 + \cdots + 5^n$, use mathematical induction to prove that

$$4S(n) = 5^{n+1} - 5 \text{ for } n \in \mathbb{N}$$ [8]

Let $P(n)$ be the statement: $4S(n) = 5^{n+1} - 5$ for all $n \in \mathbb{N}$.

Looking at $P(1)$:

L.H.S. $= 4S(1)$

$= 4(5)$

$= 20$

R.H.S. $= 5^{1+1} - 5$

$= 5^2 - 5$

$= 25 - 5$

$= 20$

Since L.H.S. = R.H.S., $P(1)$ is true.

Assume $P(k)$ is true.

That is, $4S(k) = 5^{k+1} - 5$

Looking at $P(k+1)$.

That is, $4S(k+1) = 5^{(k+1)+1} - 5$

$\Rightarrow 4S(k+1) = 5^{k+2} - 5$

We need to prove $P(k+1)$.

Taking L.H.S:

$$4S(k+1) = 4(5 + 5^2 + 5^3 + 5^4 + \cdots + 5^{k+1})$$

$$= 4(5 + 5^2 + 5^3 + 5^4 + \cdots + 5^k + 5^{k+1})$$

$$= 4[S(k) + 5^{k+1}]$$

$$= 4S(k) + 4 \cdot 5^{k+1}$$

$$= 5^{k+1} - 5 + 4 \cdot 5^{k+1}$$

$$= 5 \cdot 5^{k+1} - 5$$

$$= 5^{1+k+1} - 5$$

$$= 5^{k+2} - 5$$

$$= \text{R.H.S.}$$

$\therefore$ If $P(k)$ is true, then $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$4S(n) = 5^{n+1} - 5$ for all $n \in \mathbb{N}$.

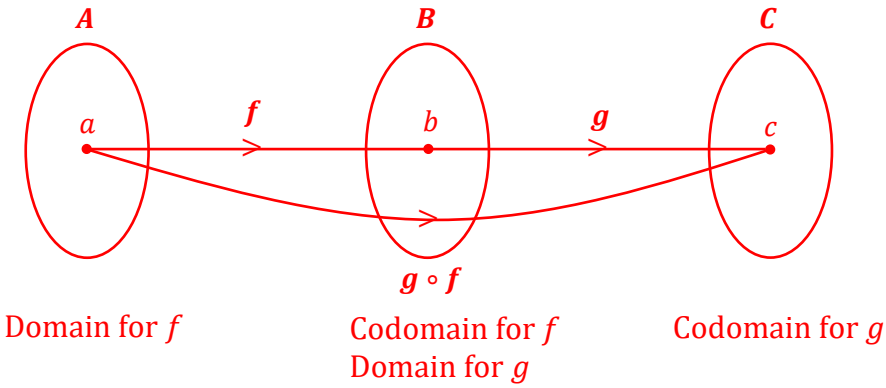
Q.E.D.

Total: 25 marks

June 2015 – Question 2 – Solution

(a) The relations $f: A \to B$ and $g: B \to C$ are functions which are both one-to-one and onto. Show that $(g \circ f)$ is

(i) one-to-one [4]



Each element of A is mapped onto one element in B and each element of B is associated with only one element in A since f is one-to-one.

$\therefore (g \circ f)$ is one-to-one.

Q.E.D.

(ii) onto [4]

The codomain of B is equal to the range since f is onto, similarly, for the function B which is mapped onto C.

$\therefore (g \circ f)$ is onto.

Q.E.D.

(b) Solve EACH of the following equations:

(i) $3 - \frac{4}{(9)^x} - \frac{4}{(81)^x} = 0$ [7]

$3 - \frac{4}{(9)^x} - \frac{4}{(81)^x} = 0$

$3 - \frac{4}{(9^x)} - \frac{4}{(9^x)^2} = 0$

Let $u = 9^x$.

Then,

$3 - \frac{4}{u} - \frac{4}{u^2} = 0$

$(\times u^2)$

$3u^2 - 4u - 4 = 0$

$(3u + 2)(u - 2) = 0$

Either $3u + 2 = 0$ or $u - 2 = 0$

$u = -\frac{2}{3}$ $u = 2$

When $u = -\frac{2}{3}$

$9^x = -\frac{2}{3}$

Taking logs on both sides:

$\log(9^x) = \log\left(-\frac{2}{3}\right)$

$\Rightarrow$ No real solutions exist for x

When $u = 2$

$9^x = 2$

Taking logs on both sides:

$\log(9^x) = \log\left(-\frac{2}{3}\right)$

$x = \frac{\log 2}{\log 9}$

$\therefore x = \frac{\log 2}{\log 9}$

(ii) $|5x - 6| = x + 5$ [5]

$|5x - 6| = x + 5$

By definition,

$5x - 6 = x + 5$ or $-(5x - 6) = x + 5$

$5x - x = 6 + 5$ $\quad -5x + 6 = x + 5$

$4x = 11$ $\quad x + 5x = 6 - 5$

$x = \frac{11}{4}$ $\quad 6x = 1$

$x = \frac{1}{6}$

Checking the solutions:

When $x = \frac{11}{4}$,

L.H.S. $= \left|5\left(\frac{11}{4}\right) - 6\right|$

$= \left|\frac{55}{4} - 6\right|$

$= \frac{31}{4}$

R.H.S. $= \frac{11}{4} + 5$

$= \frac{31}{4}$

When $x = \frac{1}{6}$

L.H.S. $= \left|5\left(\frac{1}{6}\right) - 6\right|$

$= \left|\frac{5}{6} - 6\right|$

$= \left|-\frac{31}{6}\right|$

$= \frac{31}{6}$

R.H.S. $= \frac{1}{6} + 5$

$= \frac{31}{6}$

$\therefore x = \frac{11}{4}, x = \frac{1}{6}$

(c) The population growth of bacteria present in a river after time, t hours, is given by

$$N = 300 + 5^t$$

Determine

(i) the number of bacteria present at $t = 0$ [1]

At $t = 0$,

$N = 300 + 5^0$

$= 300 + 1$

$= 301$

(ii) the time required to triple the number of bacteria [4]

Triple the bacteria $= 3 \times 301$

$= 903$

$N = 300 + 5^t$

Substituting $N = 903$ gives,

$903 = 300 + 5^t$

$5^t = 903 - 300$

$5^t = 603$

$t \ln 5 = \ln 603$

$t = \frac{\ln 603}{\ln 5}$

$= 3.977$ (to 3 d.p.)

$= 4$ hours

Total: 25 marks

June 2015 – Question 3 – Solution

(a) (i) Show that $\cos 3x = 4\cos^3 x - 3\cos x$. [6]

RTS that $\cos 3x = 4\cos^3 x - 3\cos x$

Taking LHS:

$\cos 3x = \cos(2x + x)$

$= \cos 2x \cos x - \sin 2x \sin x$ [addition formulae]

$= (2\cos^2 x - 1)\cos x - (2\sin x \cos x)\sin x$

$= 2\cos^3 x - \cos x - 2\sin^2 x \cos x$

$= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x$

$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x$

$= 4\cos^3 x - 3\cos x$

$=$ R.H.S.

$\therefore \cos 3x = 4\cos^3 x - 3\cos x$

Q.E.D.

(ii) Hence, or otherwise, solve

$$\cos 6x - \cos 2x = 0 \text{ for } 0 \le x \le 2\pi. \qquad [9]$$

$$\cos 6x - \cos 2x = 0$$

$$\cos(3 \cdot 2x) - \cos 2x = 0$$

Since $\cos 3x = 4\cos^3 x - 3\cos x$, then

$$4\cos^3 2x - 3\cos 2x - \cos 2x = 0$$

$$4\cos^3 2x - 4\cos 2x = 0$$

$$4\cos 2x\,(\cos^2 2x - 1) = 0$$

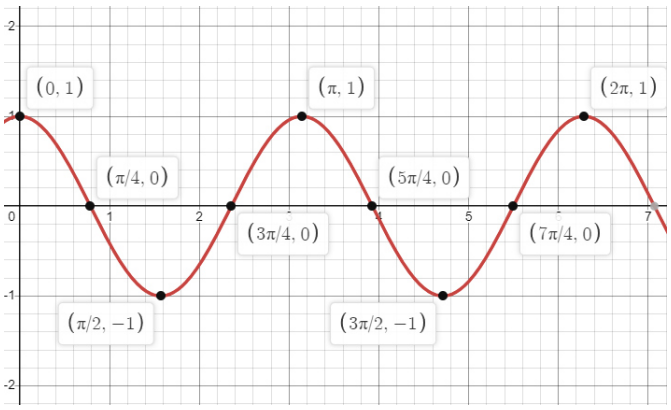
$$(4\cos 2x)(\cos 2x + 1)(\cos 2x - 1) = 0$$

When $4\cos 2x = 0$

$$\cos 2x = 0$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$



When $\cos 2x + 1 = 0$

$$\cos 2x = -1$$

$$2x = \pi, 3\pi$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

When $\cos 2x - 1 = 0$

$$\cos 2x = 1$$

$$2x = 0\,, 2\pi\,, 4\pi$$

$$x = 0\,, \pi\,, 2\pi$$

$$\therefore x = 0, \frac{\pi}{4}\,, \frac{\pi}{2}\,, \frac{3\pi}{4}\,, \pi\,, \frac{5\pi}{4}\,, \frac{3\pi}{2}\,, \frac{7\pi}{4}\,, 2\pi$$

(b) (i) Express $f(2\theta) = 3\sin 2\theta + 4\cos 2\theta$ in the form $r\sin(2\theta + \alpha)$ where

$r > 0$ and $0 < \alpha < \frac{\pi}{2}$. [6]

$$f(2\theta) = 3\sin 2\theta + 4\cos 2\theta$$

Let $3\sin 2\theta + 4\cos 2\theta = r\sin(2\theta + \alpha)$

$$= r\sin 2\theta\cos\alpha + r\cos 2\theta\sin\alpha$$

Equating coefficients gives,

$r\cos\alpha = 3 \quad \rightarrow$ Equation 1

$r\sin\alpha = 4 \quad \rightarrow$ Equation 2

Equation 2 ÷ Equation 1 gives:

$$\frac{r\sin\alpha}{r\cos\alpha} = \frac{4}{3}$$

$$\tan\alpha = \frac{4}{3}$$

$$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\alpha = 0.927^c$$

Now,

$$r = \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5$$

$$\therefore 3\sin 2\theta + 4\cos 2\theta = 5\sin(2\theta + 0.927^c)$$

(ii) Hence, or otherwise, find the maximum and minimum values of $\frac{1}{7-f(2\theta)}$. [4]

$$\frac{1}{7-f(2\theta)} = \frac{1}{7-(3\sin 2\theta + 4\cos 2\theta)}$$

$$= \frac{1}{7-5\sin(2\theta+0.927^c)}$$

Now,

$$-1 \leq \sin\theta \leq 1$$

$$-1 \leq \sin(2\theta) \leq 1$$

$$-1 \leq \sin(2\theta + 0.927^c) \leq 1$$

$$5 \geq -5\sin(2\theta + 0.927^c) \geq -5$$

$$-5 \leq -5\sin(2\theta + 0.927^c) \leq 5$$

$$7 - 5 \leq 7 - 5\sin(2\theta + 0.927^c) \leq 7 + 5$$

$$2 \leq 7 - 5\sin(2\theta + 0.927^c) \leq 12$$

$$\frac{1}{2} \geq \frac{1}{7 - 5\sin(2\theta + 0.927^c)} \geq \frac{1}{12}$$

$$\frac{1}{12} \leq \frac{1}{7 - 5\sin(2\theta + 0.927^c)} \leq \frac{1}{2}$$

Hence,

The maximum value of $\frac{1}{7 - f(2\theta)}$ is $\frac{1}{2}$.

The minimum value of $\frac{1}{7 - f(2\theta)}$ is $\frac{1}{12}$.

Total: 25 marks

June 2015 – Question 4 – Solution

(a) The circles C_1 and C_2 are defined by the parametric equations as follows:

C_1: $x = \sqrt{10}\cos\theta - 3$; $y = \sqrt{10}\sin\theta + 2$

C_2: $x = 4\cos\theta + 3$; $y = 4\sin\theta + 2$

(i) Determine the Cartesian equations of C_1 and C_2 in the form

$(x-a)^2 + (y-b)^2 = r^2$. [4]

For C_1:

$$x = \sqrt{10}\cos\theta - 3$$

$$\cos\theta = \frac{x+3}{\sqrt{10}}$$

$$y = \sqrt{10}\sin\theta + 2$$

$$\sin\theta = \frac{y-2}{\sqrt{10}}$$

Now,

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\frac{x+3}{\sqrt{10}}\right)^2 + \left(\frac{y-2}{\sqrt{10}}\right)^2 = 1$$

$$\frac{(x+3)^2}{10} + \frac{(y-2)^2}{10} = 1$$

$$\frac{(x+3)^2+(y-2)^2}{10} = 1$$

$$(x+3)^2 + (y-2)^2 = 10$$

which is of the form $(x-a)^2 + (y-b)^2 = r^2$ where $a = -3$, $b = 2$ and $r = \sqrt{10}$.

For C_2:

$$x = 4\cos\theta + 3$$

$$\cos\theta = \frac{x-3}{4}$$

$$y = 4\sin\theta + 2$$

$$\sin\theta = \frac{y-2}{4}$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\frac{x-3}{4}\right)^2 + \left(\frac{y-2}{4}\right)^2 = 1$$

$$\frac{(x-3)^2}{16} + \frac{(y-2)^2}{16} = 1$$

$$(x-3)^2 + (y-2)^2 = 16$$

which is of the form $(x-a)^2 + (y-b)^2 = r^2$ where $a = 3$, $b = 2$ and $r = 4$.

(ii) Hence, or otherwise, find the points of intersection of C_1 and C_2. [9]

C_1: $(x+3)^2 + (y-2)^2 - 10 = 0$

C_2: $(x-3)^2 + (y-2)^2 - 16 = 0$

Equating C_1 and C_2 gives:

$$(x+3)^2+(y-2)^2-10=(x-3)^2+(y-2)^2-16$$

$$(x+3)^2-10=(x-3)^2-16$$

$$x^2+6x+9-10=x^2-6x+9-16$$

$$12x=10-16$$

$$12x=-6$$

$$x=\frac{-6}{12}$$

$$x=-\frac{1}{2}$$

When $x=-\frac{1}{2}$,

$$\left(-\frac{1}{2}+3\right)^2+(y-2)^2=10$$

$$\frac{25}{4}+y^2-4y+4-10=0$$

$$y^2-4y+\frac{1}{4}=0$$

which is of the form $ay^2+by+c=0$, where $a=1, b=-4$ and $c=\frac{1}{4}$.

Using the quadratic formula:

$$y=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$=\frac{-(-4)\pm\sqrt{(-4)^2-4(1)\left(\frac{1}{4}\right)}}{2(1)}$$

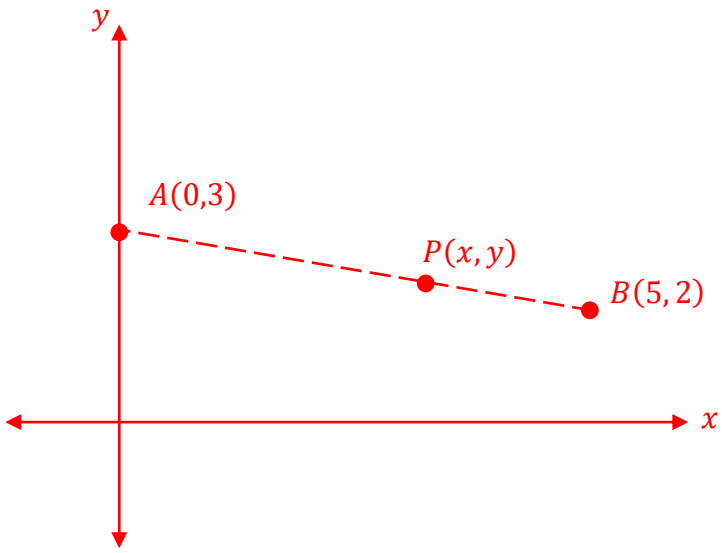
$$=\frac{4\pm\sqrt{15}}{2}$$

$$=2\pm\frac{\sqrt{15}}{2}$$

$\therefore$ The points of intersection are $\left(-\frac{1}{2}, 2+\frac{\sqrt{15}}{2}\right)$ and $\left(-\frac{1}{2}, 2-\frac{\sqrt{15}}{2}\right)$.

(b) A point $P(x, y)$ moves so that its distance from the fixed point $(0, 3)$ is two times the distance from the fixed point $(5, 2)$. Show that the equation of the locus of the point $P(x, y)$ is a circle. [12]

A point $P(x, y)$ moves so that its distance from the fixed point $(0, 3)$ is two times the distance from the fixed point $(5, 2)$.



$$|PA| = 2|PB|$$

$$\sqrt{(x-0)^2 + (y-3)^2} = 2\sqrt{(x-5)^2 + (y-2)^2}$$

Squaring both sides:

$$x^2 + (y-3)^2 = 4[(x-5)^2 + (y-2)^2]$$

$$x^2 + y^2 - 6y + 9 = 4(x^2 - 10x + 25 + y^2 - 4y + 4)$$

$$x^2 + y^2 - 6y + 9 = 4x^2 - 40x + 100 + 4y^2 - 16y + 16$$

$$3x^2 + 3y^2 - 40x - 10y + 107 = 0$$

$(\div 3)$

$x^2 + y^2 - \frac{40}{3}x - \frac{10}{3}y + \frac{107}{3} = 0$

which is of the form $x^2 + y^2 + 2fx + 2gy + c = 0$

where $f = -\frac{20}{3}$, $g = -\frac{5}{3}$ and $c = \frac{107}{3}$

This is the equation of a circle with centre $\left(\frac{20}{3}, \frac{5}{3}\right)$ and radius,

$r = \sqrt{\left(-\frac{20}{3}\right)^2 + \left(-\frac{5}{3}\right)^2 - \frac{107}{3}} = \frac{104}{9}$ units.

Q.E.D.

Total: 25 marks

June 2015 – Question 5 – Solution

(a) Let f be a function define as

$$f(x) = \begin{cases} \frac{\sin(ax)}{x} & if\ x \neq 0, \quad a \neq 0 \\ 4 & if\ x = 0 \end{cases}$$

If f is continuous at $x = 0$, determine the value of a. [4]

Since $f(x)$ is continuous at $x = 0$, then

$$\lim_{x \to 0} f(x) = f(0)$$

$$\lim_{x \to 0} \frac{\sin(ax)}{x} = 4$$

$$\lim_{x \to 0} \frac{\sin(ax)}{x} \cdot \frac{a}{a} = 4$$

$$a \lim_{x \to 0} \frac{\sin(ax)}{ax} = 4$$

$$a(1) = 4$$

$$a = 4$$

$\therefore$ The value of $a = 4$.

(b) **Using first principles**, determine the derivative of $f(x) = \sin(2x)$. [6]

Let $f(x) = \sin 2x$.

Using first principles,

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\sin[2(x+h)]-\sin 2x}{h}$$

$$= \lim_{h\to 0} \frac{\sin(2x+2h)-\sin 2x}{h}$$

$$= \lim_{h\to 0} \frac{2\cos\left(\frac{2x+2h+2x}{2}\right)\sin\left(\frac{2x+2h-2x}{2}\right)}{h}$$

$$= \lim_{h\to 0} \frac{2\cos(2x+h)\sin(h)}{h}$$

$$= 2\lim_{h\to 0}\cos(2x+h) \times \lim_{h\to 0}\frac{\sin h}{h}$$

$$= 2\cos(2x) \times 1$$

$$= 2\cos 2x$$

$\therefore$ The derivative is $f'(x) = 2\cos 2x$.

(c) If $y = \frac{2x}{\sqrt{1+x^2}}$ show that

(i) $x\frac{dy}{dx} = \frac{y}{1+x^2}$ [7]

$$y = \frac{2x}{\sqrt{1+x^2}}$$

$$y = 2x(1+x^2)^{-\frac{1}{2}}$$

which is in the form $y = uv$.

Let $u = 2x$ and $v = (1+x^2)^{-\frac{1}{2}}$

$\frac{du}{dx} = 2$ $\frac{dv}{dx} = -\frac{1}{2}(2x)(1+x^2)^{-\frac{3}{2}}$

$\frac{dv}{dx} = -x(1+x^2)^{-\frac{3}{2}}$

Using the quotient rule,

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$= (2x)(-x)(1+x^2)^{-\frac{3}{2}} + (1+x^2)^{-\frac{1}{2}}(2)$$

$$= -2x^2(1+x^2)^{-\frac{3}{2}} + 2(1+x^2)^{-\frac{1}{2}}$$

Required to show that $x\frac{dy}{dx} = \frac{y}{1+x^2}$.

Taking L.H.S:

$$x\frac{dy}{dx} = x \cdot \left[-2x^2(1+x^2)^{-\frac{3}{2}} + 2(1+x^2)^{-\frac{1}{2}}\right]$$

$$= -2x^3(1+x^2)^{-\frac{3}{2}} + 2x(1+x^2)^{-\frac{1}{2}}$$

$$= 2x(1+x^2)^{-\frac{3}{2}}[-x^2 + (1+x^2)^1]$$

$$= 2x(1+x^2)^{-\frac{3}{2}}(1)$$

$$= 2x(1+x^2)^{-\frac{1}{2}}(1+x^2)^{-1}$$

$$= \frac{2x(1+x^2)^{-\frac{1}{2}}}{(1+x^2)}$$

$$= \frac{y}{(1+x^2)}$$

$$= \text{R.H.S.}$$

$\therefore x\frac{dy}{dx} = \frac{y}{(1+x^2)}$ Q.E.D.

(ii) $\frac{d^2y}{dx^2} + \frac{3y}{(1+x^2)^2} = 0$ [8]

From previous,

$x\frac{dy}{dx} = 2x(1+x^2)^{-\frac{3}{2}}(1)$

$(\div x)$

$\frac{dy}{dx} = 2(1+x^2)^{-\frac{3}{2}}$

Now,

$\frac{d^2y}{dx^2} = 2(-\frac{3}{2})(2x)(1+x^2)^{-\frac{5}{2}}$

$= -6x(1+x^2)^{-\frac{5}{2}}$

Required to prove that $\frac{d^2y}{dx^2} + \frac{3y}{(1+x^2)^2} = 0$.

Taking L.H.S:

$\frac{d^2y}{dx^2} + \frac{3y}{(1+x^2)^2} = -6x(1+x^2)^{-\frac{5}{2}} + \frac{3y}{(1+x^2)^2}$

$= -\frac{3\left[2x(1+x^2)^{-\frac{1}{2}}\right]}{(1+x^2)^2} + \frac{3y}{(1+x^2)^2}$

$= -\frac{3y}{(1+x^2)^2} + \frac{3y}{(1+x^2)^2}$

$= 0$

$=$ R.H.S.

$\therefore \frac{d^2y}{dx^2} + \frac{3y}{(1+x^2)^2} = 0$

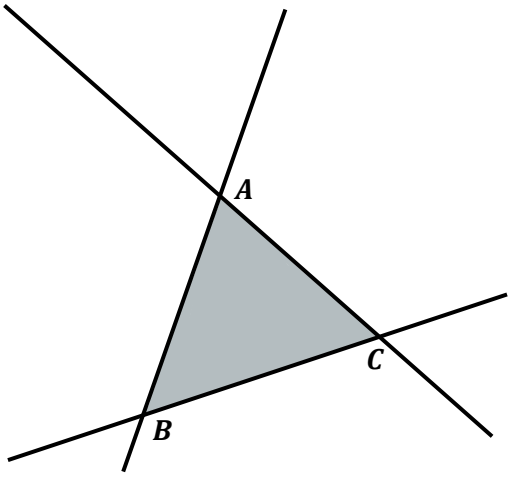
Q.E.D.

Total: 25 marks

June 2015 – Question 6 – Solution

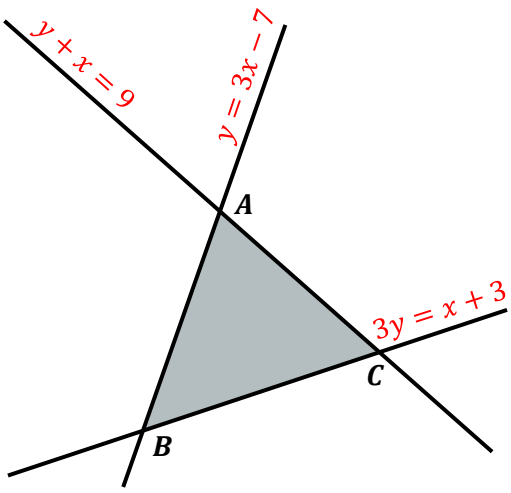
(a) The diagram below (**not drawn to scale**) shows the region bounded by the lines

$$y = 3x - 7, \quad y + x = 9 \quad \text{and} \quad 3y = x + 3.$$



(i) Show that the coordinates of A, B and C are $(4, 5)$, $(3, 2)$, and $(6, 3)$ respectively. [5]

We can label the lines as follows:



$y = 3x - 7 \qquad \rightarrow$ Equation 1

$y + x = 9 \qquad \rightarrow$ Equation 2

Substituting Equation 1 into Equation 2 gives:

$3x - 7 + x = 9$

$3x + x = 9 + 7$

$4x = 16$

$x = \frac{16}{4}$

$x = 4$

Substituting $x = 4$ into Equation 1 gives:

$y = 3(4) - 7$

$= 12 - 7$

$= 5$

So, we have the point $A(4, 5)$.

$3y = x + 3 \qquad \rightarrow$ Equation 1

$y = 3x - 7 \qquad \rightarrow$ Equation 2

Substituting Equation 2 into Equation 1 gives:

$3(3x - 7) = x + 3$

$9x - 21 = x + 3$

$9x - x = 21 + 3$

$8x = 24$

$x = \frac{24}{8}$

$x = 3$

Substituting $x = 3$ into Equation 2 gives:

$y = 3(3) - 7$

$= 9 - 7$

$= 2$

So, we have the point $B(3, 2)$.

$3y = x + 3$ $\rightarrow$ Equation 1

$y + x = 9$ $\rightarrow$ Equation 2

Rearranging Equation 2 gives:

$y = 9 - x$ $\rightarrow$ Equation 3

Substituting Equation 3 into Equation 1 gives:

$3(9 - x) = x + 3$

$27 - 3x = x + 3$

$27 - 3 = 3x + x$

$24 = 4x$

$x = \frac{24}{4}$

$x = 6$

Substituting $x = 6$ into Equation 3 gives:

$y = 9 - 6$

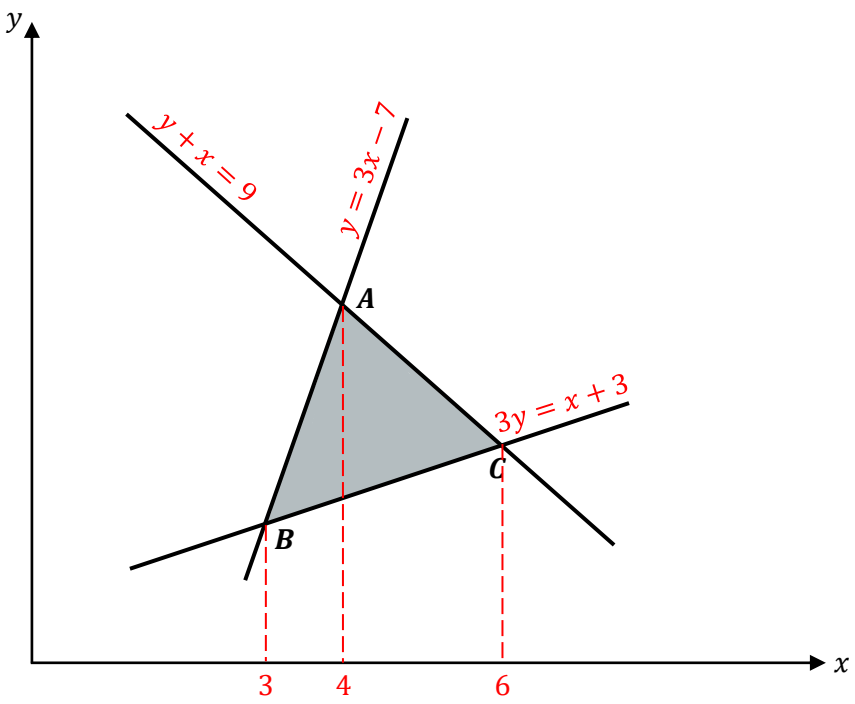
$= 3$

So, we have the point $C(6, 3)$.

$\therefore$The coordinates are $A(4, 5)$, $B(3, 2)$ and $C(6, 3)$. Q.E.D.

(ii) Hence, use integration to determine the area bounded by the lines. [6]

Consider the diagram below:



Area of shaded region = Area under AB + Area under AC − Area under BC

Area under $AB = \int_a^b y\,dx$

$= \int_3^4 (3x - 7)\,dx$

$= \left[\frac{3x^2}{2} - 7x\right]_3^4$

$= \left[\frac{3(4)^2}{2} - 7(4)\right] - \left[\frac{3(3)^2}{2} - 7(3)\right]$

$= 24 - 28 - \left(\frac{27}{2} - 21\right)$

$= 24 - 28 + \frac{15}{2}$

$= \frac{7}{2}$ units^2

$y + x = 9$

$y = 9 - x$

Area under $AC = \int_a^b y\,dx$

$= \int_4^6 (9 - x)\,dx$

$= \left[9x - \frac{x^2}{2}\right]_4^6$

$= \left[9(6) - \frac{(6)^2}{2}\right] - \left[9(4) - \frac{(4)^2}{2}\right]$

$= 54 - 18 - (36 - 8)$

$= 54 - 18 - 28$

$= 8 \text{ units}^2$

$3y = x + 3$

$y = \frac{1}{3}x + 1$

Area under $BC = \int_a^b y\,dx$

$= \int_3^6 \left(\frac{1}{3}x + 1\right) dx$

$= \left[\frac{x^2}{6} + x\right]_3^6$

$= \left[\frac{(6)^2}{6} + (6)\right] - \left[\frac{(3)^2}{6} + (3)\right]$

$= 6 + 6 - \left(\frac{3}{2} + 3\right)$

$= 6 + 6 - \frac{9}{2}$

$= \frac{15}{2} \text{ units}^2$

Now,

Area of shaded region = Area under AB + Area under AC − Area under BC

$$= \frac{7}{2} + 8 - \frac{15}{2}$$

$$= 4 \text{ units}^2$$

(b) The gradient function of a curve $y = f(x)$ which passes through the point $(0, -6)$ is given by $3x^2 + 8x - 3$.

(i) Determine the equation of the curve. [3]

$$\frac{dy}{dx} = 3x^2 + 8x - 3$$

$$y = \int \frac{dy}{dx}\, dx$$

$$y = \int (3x^2 + 8x - 3)\, dx$$

$$y = \frac{3x^3}{3} + \frac{8x^2}{2} - 3x + c$$

$$y = x^3 + 4x^2 - 3x + c \qquad \text{(G.S.)}$$

Substituting $(0, -6)$ into the G.S. gives:

$$-6 = (0)^3 + 4(0)^2 - 3(0) + c$$

$$-6 = 0 + 0 - 0 + c$$

$$c = -6$$

Substituting $c = -6$ into the G.S. gives:

$$y = x^3 + 4x^2 - 3x - 6 \qquad \text{(P.S.)}$$

∴ The equation of the curve is $y = x^3 + 4x^2 - 3x - 6$.

(ii) Find the coordinates and nature of the stationary point of the curve in (b)(i) above. [8]

$\frac{dy}{dx} = 3x^2 + 8x - 3$

At stationary points, $\frac{dy}{dx} = 0$.

When $\frac{dy}{dx} = 0$,

$3x^2 + 8x - 3 = 0$

$(3x - 1)(x + 3) = 0$

Either $3x - 1 = 0$ or $x + 3 = 0$

$3x = 1$ $x = -3$

$x = \frac{1}{3}$

When $x = \frac{1}{3}$,

$y = \left(\frac{1}{3}\right)^3 + 4\left(\frac{1}{3}\right)^2 - 3\left(\frac{1}{3}\right) - 6$

$= \frac{1}{27} + 4\left(\frac{1}{9}\right) - 1 - 6$

$= \frac{1}{27} + \frac{4}{9} - 1 - 6$

$= -\frac{176}{27}$

Therefore, the stationary point is $\left(\frac{1}{3}, -\frac{176}{27}\right)$.

When $x = -3$,

$y = (-3)^3 + 4(-3)^2 - 3(-3) - 6$

$= -27 + 4(9) + 9 - 6$

$= 12$

Therefore, the stationary point is $(-3, 12)$.

Now,

$\frac{dy}{dx} = 3x^2 + 8x - 3$

$\frac{d^2y}{dx^2} = 6x + 8$

When $x = \frac{1}{3}$,

$\frac{d^2y}{dx^2} = 6\left(\frac{1}{3}\right) + 8$

$= 10 \quad (> 0)$

Since $\frac{d^2y}{dx^2} > 0$, then $\left(\frac{1}{3}, -\frac{176}{27}\right)$ is a minimum point.

When $x = -3$,

$\frac{d^2y}{dx^2} = 6(-3) + 8$

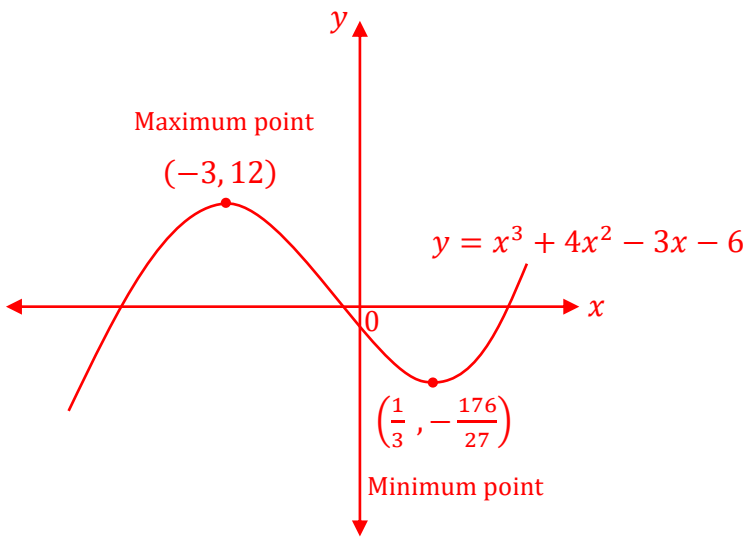
$= -10 \quad (< 0)$

Since $\frac{d^2y}{dx^2} < 0$, then $(-3, 12)$ is a maximum point.

(iii) Sketch the curve in (b)(i) by clearly labelling the stationary points. [3]

$\left(\frac{1}{3}, -\frac{176}{27}\right)$ is a minimum point and $(-3, 12)$ is a maximum point.

A sketch of the curve is drawn below:



Total: 25 marks